EXAMPLE: FIFTH-ORDER POLYNOMIAL TRAJECTORY PLANNING

FIRST JOINT OF A 6 DOF ROBOT GO FROM 30° TO A FINAL ANGLE OF 75° IN 5s. INITIAL AND FINAL ACCELERATION/DECELERATION IS 5°/s². USING A FIFTH-ORDER POLYNOMIAL, DEFINE THE POSITION, VELOCITY AND ACCELERATION FOR THIS JOINT.

SOLUTION: WE HAVE

$$\theta_i = 30^\circ \quad \dot{\theta}_i = 0^\circ/s \quad \ddot{\theta}_i = 5^\circ/s^2$$
$$\theta_F = 75^\circ \quad \dot{\theta}_F = 0^\circ/s \quad \ddot{\theta}_F = -5^\circ/s^2$$

FIFTH-ORDER EQUATION

$$\theta(t) = c_0 + c_1 t + c_2 t^2 + c_3 t^3 + c_4 t^4 + c_5 t^5$$

VELOCITY: $\dot{\theta}(t) = c_1 + 2c_2 t + 3c_3 t^2 + 4c_4 t^3 + 5c_5 t^4$

ACCELERATION: $\ddot{\theta}(t) = 2c_2 + 6c_3 t + 12c_4 t^2 + 20c_5 t^3$

FOR INITIAL CONDITIONS

$$\theta_i = \theta(0) = c_0 = 30$$
$$\dot{\theta}_i = \dot{\theta}(0) = c_1 = 0$$
$$\ddot{\theta}_i = \ddot{\theta}(0) = 2c_2 = 5 \qquad \therefore c_2 = 2.5$$

FOR FINAL CONDITIONS

$$\theta_F = \theta(t_F) = c_0 + c_1 t_F + c_2 t_F^2 + c_3 t_F^3 + c_4 t_F^4 + c_5 t_F^5$$
$$75 = 30 + 62.5 + c_3(125) + 625 c_4 + 3125 c_5$$
$$75 = 92.5 + 125 c_3 + 625 c_4 + 3125 c_5$$
$$0 = 17.5 + 125 c_3 + 625 c_4 + 3125 c_5 \qquad -(1)$$

VELOCITY:

$$\dot{\theta}_F = \dot{\theta}(t_F) = c_1 + 2c_2 t + 3c_3 t^2 + 4c_4 t^3 + 5c_5 t^4$$

$$0 = 2c_2(5) + 3c_3(25) + 4c_4(125) + 5c_5(625)$$

$$0 = 10c_2 + 75c_3 + 500c_4 + 3125c_5$$

$$0 = 25 + 75c_3 + 500c_4 + 3125c_5 \quad (2)$$

ACCELERATION:

$$\ddot{\theta}_F = \ddot{\theta}(t_F) = 2c_2 + 6c_3 t_F + 12c_4 t_F^2 + 20c_5 t_F^3$$

$$-5 = 5 + 30c_3 + 300c_4 + 2500c_5$$

$$0 = 10 + 30c_3 + 300c_4 + 2500c_5 \quad (3)$$

REARRANGING (3) GIVES

$$c_3 = -0.333 - 10c_4 - 83.333c_5 \quad (4)$$

SUBSTITUTE (4) INTO (1) GIVES

$$c_4 = -0.039 - 11.667c_5 \quad (5)$$

SUBSTITUTE (4) INTO (2) GIVES

$$0 = 0.025 - 250c_4 - 3125c_5 \quad (6)$$

SUBSTITUTE (6) INTO (5) GIVES

$$c_5 = 0.0469 \quad (7)$$

SUBSTITUTE (7) INTO (5) GIVES

$$c_4 = -0.586 \quad (8)$$

SUBSTITUTE (7) AND (8) INTO (4) GIVES

$$c_3 = 1.619 \quad (9)$$

THESE COEFFICIENTS GIVE THE FOLLOWING FIFTH-ORDER TRAJECTORY POSITION, VELOCITY AND ACCELERATION:

$$\theta(t) = 30 + 2.5t^2 + 1.619t^3 - 0.586t^4 + 0.0469t^5$$

$$\dot{\theta}(t) = 5t + 4.857t^2 - 2.344t^3 + 0.235t^4$$

$$\ddot{\theta}(t) = 5 + 9.714t - 7.032t^2 + 0.938t^3$$